

Katharina Sternbauer

Computer Scientist

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Professional Experience



Medical IT @Red Cross Upper Austria

 January 2025 - Now  Linz, Austria)

- Medical IT with a focus on Data Warehouse, Tableau, Electronic Content Management
- Maintaining older software pieces and bringing them up-to-date
- Part-time employment, 24-35h / week



Data Engineer & Developer @Primetals Technologies

 November 2023 - September 2024  Linz, Austria

- DWH engineer to integrate adaptive ETL processes for Data Warehousing.
- Part-time employment, 24h / week



Java Developer @Primetals Technologies

 October 2020 - October 2021, February 2022 - November 2023  Linz, Austria

- Software Engineer for Continuous Casting Machines
- Working with Eclipse SWT and Java to develop software for casting machine design.
- Student employee, part-time, 10-20h / week



Network Technician @Technician at JKU

 October 2021 - February 2022  Linz, Austria

- Responsible at the Institute for Digital Business for hardware maintenance, server and software management, technical assistant for exams.
- Student employee, part-time, 10h / week

Internships @Primetals Austria, Stiwa Group and Technosert

 2014, 2015, 2016, 2019, 2020  Linz - Hagenberg - Pregarten, Austria

Education



Master of Science in Computer Science @Johannes Kepler Universität

 2021 - 2025  Linz, Austria

- Major: Data Science
- Graduation certificate: Straight A's
- Thesis: Adaptive ETL processing for Business Intelligence applications with Data Warehouses



Bachelor of Science in Computer Science @Johannes Kepler Universität

 2018 - 2022  Linz, Austria

- Thesis: Performance Evaluation of the LMS/HSS hash-based signature system
- Passed with distinction

High Technical College (HTL) @Linzer Technikum Paul-Hahn-Straße

 2012 - 2017  Linz, Austria

- Matura: Passed with distinction
- All years passed with distinction / flying colours

Featured Projects



Passion for technology @Personal

⌚ 2011 - Never ending

- Building my own PCs, which started my interest in Computer Science
- 3D design and printing since 2020
- Self-assembled and configured NAS-system on Ubuntu Server with LVM, Raid, Samba, DLNA
- Tech editor for German portal tech-review.de from 2014 until 2024
- Laying patch cables everywhere to ensure sufficient network coverage
- Student dorm network admin team from 2021 to 2024
- Raspberry Pi as a Docker host, network DNS guard, DNS-Updater service, Home-API, Time Series Databases



Adaptive ETL processing for Business Intelligence applications with Data Warehouses @Master Thesis

⌚ 2023 - 2025 📍 JKU

- My master thesis, based on the concept of automated code generation based on semantic, ontological data
- Java, MSSQL, Typst
- Publication: <https://epub.jku.at/obvulihs/content/titleinfo/11591465>



Ear Spoof Detection Neural Network @University Project

⌚ 2023 📍 JKU

- As my master project, based on a seminar work, I implemented a neural network which can distinguish genuine biometric identifiers from fakes
- The network is trained to detect spoofed data for biometric ear identification
- Python, PyTorch, Kaggle



Personal Webpage @Personal

⌚ 2022 - Today

- The source for my webpage is also freely available. It uses Bootstrap 5, directly loaded from a CDN, as its CSS framework and utilizes very little custom CSS and JavaScript for some smaller refinements.
- HTML, CSS, Javascript



AirBnB Barcelona Visualization @University Group Project

⌚ 2022 📍 JKU

- As part of the exercise track of “Visual Analytics” at JKU, we had to take a deep dive into data analysis including static plotting, correlation and getting new insights with techniques such as PCA and clustering.
- Implemented various statistical methods to assess the different pricing of AirBnB rooms in Barcelona
- Integrated Python libraries for visualization of geographical locations, graphs and plots
- Python, Jupyter Lab, Geopandas

Skills & Interests

Technology

Data Engineering

Data Warehousing

Machine Learning

Docker / Container

Scripting

PC Hardware

3D printing & design

Languages

German (native)

English (fluent)

Java / Kotlin

Rust (beginner)

C (basics)

SQL

Python (fundamentals)

C# (fundamentals)

Shell

Sports

Boulder (regular)

Rope climbing

Parkour (beginner)

Interests

Reading

Music

Politics

Social topics

PC-related tech

Privacy

Internet of Things